# Bt512 midterm solved imp mcqs 1

Microbial Biotechnology (Virtual University of Pakistan)



Scan to open on Studocu

Studocu is not sponsored or endorsed by any college or university
Downloaded by Shahzeb Mughal (shahzebvip01@gmail.com)

# BT512 Microbial Biotechnology-Full MCQs

## Section 1:Microbial Basics

1. Which type of bacteria produce lactic acid during fermentation?

- *A) Escherichia coli*
- B) *Lactobacillus*
- C) *Clostridium*
- D) *Pseudomonas*

Answer:B

2. The microbial conversion of glucose to ethanol is known as:

- A)Respiration
- B) Alcoholic fermentation
- C) Photosynthesis
- D)Glycolysis

Answer:B

3. Which of the following microbes is anaerobic?

- A) *Clostridium botulinum*
- *B)E.coli*
- C) *Saccharomyces cerevisiae*
- D) *Bacillus subtilis*

Answer:A

4. Microbial biomass production is maximized in which growth phase?

- A) Lag phase
- B)Log phase
- C) Stationary phase
- D)Death phase

Answer:B

This document is available on studocu
Downloaded by Shahzeb Mughal (shahzebvip01@gmail.com)
Scanned with CamScanner

## Section 2: Industrial Applications

5. The industrial production of vinegar uses:

- *A)Lactobacillus*
- *B)Acetobacter*
- C) *Aspergillus*
- *D) Rhizobium*

Answer:B

6. Which of the following is used for the production of glutamic acid?

- *A) Corynebacterium glutamicum*
- *B)E.coli*
- *C)Streptococcus*
- D) *Saccharomyces cerevisiae*

Answer:A

7. Microbes used for enzyme production in detergents include:

- *A)Bacillus licheniformis*
- *B) Pseudomonas fluorescens*
- *C)E.coli*
- D) *Aspergillus oryzae*

Answer:A

8. Which industry uses *Aspergillus niger for citric acid production?*

- A)Pharmaceutical
- B) Food and beverage
- C) Textile
- D) Agriculture

Answer:B

Downloaded by Shahzeb Mughal (shahzebvip01@gmail.com)

Scanned with CamScanner

## Section 3:Genetic Engineering

9. In recombinant DNA technology, restriction enzymes are known as:

- A)Molecular markers
- B) Molecular scissors
- C) Molecular clones
- D) None of the above

Answer:B

10. The first recombinant human protein produced in *E.coliwas:*

- A)Insulin
- B)Growth hormone
- C) Interferon
- D)Penicillin

Answer:A

11. Gene cloning in bacteria requires:

- A) A host, a vector, and the gene of interest
- B)An RNA template
- C) PCR only
- D) Only a plasmid

Answer:A

12. Golden rice is an example of:

- A) Industrial biotechnology
- B) Microbial remediation
- C)Genetic engineering
- D)Biofertilizer

Answer:C

This document is available on studocu
Downloaded by Shahzeb Mughal (shahzebvip01@gmail.com)
Scanned with CamScanner

## Section 4:Medical Applications

13. Which microorganism produces streptomycin?

- *A)Penicillium notatum*
- *B)Streptomyces griseus*
- *C) Bacillus subtilis*
- D) *Lactobacillus acidophilus*

Answer:B

14. The use of bacteriophages to treat bacterial infections is called:

- A) Antibiotic therapy
- B)Bioremediation
- C) Phage therapy
- D) Vaccination

Answer:C

15. Which microbe is used for producing vaccines against hepatitis B?

- A)E.coli
- *B) Saccharomyces cerevisiae*
- C) *Pseudomonas*
- D) *Bacillus subtilis*

Answer:B

16. Human insulin is produced using:

- A) *Saccharomyces cerevisiae*
- B)E.coli
- C) *Aspergillus niger*
- D) *Lactobacillus*

Answer:B

Downloaded by Shahzeb Mughal (shahzebvip01@gmail.com)
Scanned with CamScanner

## Section 5:Environmental Blotechnology

17. Which of the following microbes is used in biogas production?

- *A) Methanobacterium*
- *B)Penicillium*
- *C)E.coli*
- D) *Lactobacillus*

Answer:A

18. The cleanup of environmental pollutants by microorganisms is called:

- A)Bioaugmentation
- B)Bioremediation
- C) Composting
- D)Fermentation

Answer:B

19. Biofertilizers are often composed of:

- A) *Rhizobium*
- B) *Azospirillum*
- C) *Mycorrhiza*
- D) All of the above

Answer:D

20. Microbial fuel cells produce electricity by:

- A)Oxidizing inorganic substrates
- B) Reducing pollutants
- C) Direct electron transfer by bacteria
- D) Generating methane gas

Answer:C

This document is available on studocu
Downloaded by Shahzeb Mughal (shahzebvip01@gmail.com)
Scanned with CamScanner

## Section 6:Advanced Microbial Techniques

21. PCR is used to:

- A) Cut DNA
- B)Amplify DNA

C)Sequence DNA

- D) Clone cells

Answer:B

22. Which type of sequencing method uses fluorescent tags?

- A)Sanger sequencing
- B) Shotgun sequencing
- C)PCR
- D) CRISPR analysis

Answer:A

23. Which technique uses microbes to extract metals from ores?

- A)Bioleaching
- B)Biosorption
- C)Composting
- D)Fermentation

Answer:A

Downloaded by Shahzeb Mughal (shahzebvip01@gmail.com)

Scanned with CamScanner CamScanner